01219231: Database Systems for Software and Knowledge Engineers (Tentative)

Semester: 2026 / 1st

Instructors:

- Dr. Kitsana Waiyamai (kitsana.w@ku.th)

- Dr. Piya Limcharoen (piya.lim@ku.th)

- TA: P'Perth (thanawas.s@ku.th)

Class Time: Tuesdays 13:00–16:00, Room s601

Google Classroom: code of5lsbaf (use your @ku.th account)

Course Description (Updated)

This course provides students with a solid foundation in modern database systems, covering both classical and contemporary topics. Topics include data modeling, entity-relationship (ER) and UML models, relational model and relational algebra, SQL and query processing, constraints, views, indexes, normalization, transaction management, concurrency control, NoSQL models (document, key-value, columnar, and graph), modern databases (NewSQL, time series, multi-model), and emerging trends such as cloud databases, real-time analytics, vector databases, and database applications with modern tools.

Learning Outcomes

Upon successful completion of this course, students will be able to:

• Explain the key components and architectures of modern DBMS.
• Design conceptual and logical schemas for relational and NoSQL systems.
• Query and manipulate data using SQL and NoSQL query languages.
• Normalize database schemas and apply design principles.
• Evaluate various types of databases (e.g., relational, NoSQL, cloud, distributed) for different applications. • Implement and deploy database applications with appropriate tools.

Evaluation

• 20% Homework, Quizzes, Class Participation
• 20% Group Project
• 30% Midterm Exam
• 30% Final Exam

Course Schedule (Updated)

Week	Date	Topic	Instructor
1	Aug 04	Introduction to Databases (Trends and Types of DBs)	Kitsana
2	Aug 11	Data Models: Relational, NoSQL, and Beyond (Document, Graph,	Kitsana

		Key Value)	
3	Aug 18	Relational Model, ER to Relational Mapping	Kitsana
4	Aug 25	SQL Basics: DDL, DML, Constraints	Kitsana
5	Sep 01	SQL Advanced: Joins, Views, Aggregates, Indexes	Kitsana
6	Sep 08	UML for Database Design Theory	Kitsana
7	Sep 15	Normalization and Design Theory	Kitsana
—	Sept 19-27	Midterm Examination	—
8	Sep	Transactions, Concurrency, Recovery	Piya
—	Oct	No Class (Commencement Week)	—
9	Oct	Developing Applications with Relational DBs (e.g., PostgreSQL + Python)	Piya
10	Oct	Query Processing and Optimization	Piya
11	Oct	Distributed DBs and NoSQL Systems (CAP Theorem, BASE, etc.)	Piya
12	Nov	MongoDB (Document Model + Aggregation Framework)	Piya
13	Nov	MapReduce & Big Data Pipelines (e.g., Hadoop, Spark SQL)	Piya
14	Nov	Modern Topics: Cloud DBs, Vector DBs, Real-Time, Serverless	Piya

15	Nov	Group Project Presentation	Piya

—	Dec	Final Examination	—